

A NOTE ON BAYESIAN D-OPTIMAL DESIGNS FOR A GENERALIZATION OF THE EXPONENTIAL GROWTH MODEL

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Summary: In this note the problem of constructing Bayesian D-optimal designs in a two-parameter nonlinear regression model is considered which generalizes an exponential growth model. Optimal two-point designs are determined analytically and it is shown that these designs are also Bayesian D-optimal in the class of all designs provided a gamma distribution with sufficiently large parameter value is used as prior distribution. Numerical results for Bayesian D-optimal three and (if necessary) four-point designs are presented if the best two-point designs are not globally optimal.

1. INTRODUCTION

The construction of optimal Bayesian experimental designs for two-parameter nonlinear growth models has been discussed intensively in the recent literature. Some general aspects in the theory of Bayesian optimal designs have been investigated by Chaloner (1987) and Chaloner and Larntz (1989, 1992). Mukhopadhyay and Haines (1994) and Dette and Neugebauer (1994) demonstrate the difficulties appearing in the construction of Bayesian D-optimal designs even in the relatively simple exponential growth model. Among other things it is shown in these papers that (approximate) two-point designs are Bayesian D-optimal for the exponential growth model if a

gamma distribution with sufficiently large parameter is assumed for prior for the nonlinear parameter in the model.

This note is intended to serve as an addendum to the results established recently in Mukhopadhyay and Haines (1994) and Dette and Neugebauer (1994). In section 2 we will consider a more general nonlinear model than in these references such that the response function has a maximum. The simple exponential growth model appears as a limiting case. We determine the optimal two-point designs and show that these designs are also optimal within the class of all designs if gamma distributions with sufficiently large parameters are used as prior. If these parameters are too small the two-point design is not Bayesian D-optimal. In section 3 some numerical results for optimal three- and (if necessary) four-point designs are presented and it is demonstrated that the number of support points is increasing with the variance of the prior distribution.

2. A NONLINEAR MODEL

Throughout this paper we use the usual notation of approximate design theory for nonlinear models. For a more complete introduction we refer the reader to the work of Chaloner (1987) and Chaloner and Larntz (1989, 1992). We will consider the nonlinear regression model

$$(1) \quad y = f(x, \theta) + \epsilon = \vartheta_1 x^k \exp(-\vartheta_2 x) + \epsilon, \quad x \geq 0$$

where $\theta = (\vartheta_1, \vartheta_2)'$, $\vartheta_1, \vartheta_2 > 0$ denotes the unknown vector of parameters, $k \geq 0$ is a known constant and $\epsilon = \epsilon(x)$ is a normally distributed error term with mean 0 and variance σ^2 . We assume further that the observations taken for the estimation of θ are independent. In the case $k = 0$ this model reduces to the exponential regression model considered by Mukhopadhyay and Haines (1994) and Dette and Neugebauer (1994). For a single observation at $x \geq 0$ the Fisher information for θ in the model (1) is given by the matrix

$$I(\theta, x) = \left[\frac{\partial}{\partial \theta} f(x, \theta) \right] \left[\frac{\partial}{\partial \theta} f(x, \theta) \right]^T = \begin{bmatrix} x^{2k} e^{-2\vartheta_2 x} & -\vartheta_1 x^{2k+1} e^{-2\vartheta_1 x} \\ -\vartheta_1 x^{2k+1} e^{-2\vartheta_2 x} & \vartheta_1^2 x^{2k+2} e^{-2\vartheta_1 x} \end{bmatrix}$$

where $\frac{\partial}{\partial \theta} f(x, \theta)$ denotes the gradient of f . An approximate design is a probability measure ξ on the positive half line $[0, \infty)$ with finite support and the matrix

$$I(\theta, \xi) = \int_0^\infty I(\theta, x) d\xi(x)$$

is called the information matrix of ξ . Throughout this paper designs with p support points are called p -point designs ($p \in \mathbb{N}$). If ξ puts masses $\frac{n_j}{N}$ at x_j ($j = 1, \dots, p$), then N independent observations are taken, n_j at each x_j , in order to estimate the unknown parameter θ . In this case the matrix $I(\theta, \xi)$ is proportional to the inverse of the asymptotic covariance matrix of the maximum likelihood estimator for θ .

Let π denote a prior distribution for $\theta = (\vartheta_1, \vartheta_2)'$ with independent marginal distributions π_1 and π_2 . A design ξ^* is called Bayesian D-optimal (with respect to the prior π) if ξ^* maximizes

$$\Phi_\pi(\xi) = E_\pi[\log(\det I(\theta, \xi))] = \int \log(\det I(\theta, \xi)) d\pi(\theta)$$

and $\Phi_\pi(\xi^*) < \infty$. Here $E_\pi[\cdot]$ denotes the expectation with respect to the prior distribution π and we assume that this prior is sufficiently regular such that there exists at least one design ξ with $E_\pi[|\log(\det I(\theta, \xi))|] < \infty$. On the model (1) a simple calculation shows

$$E_\pi[\log(\det I(\theta, \xi))] = 2E_\pi[\log(\vartheta_1)] + E_\pi[\log(\det I(\theta^*, \xi))]$$

where $\theta^* = (1, \vartheta_2)'$. Consequently the Bayesian D-optimal design depends on the prior distribution π only through the marginal distribution π_2 of ϑ_2 . The following proposition gives the best Bayesian D-optimal design in

the class of all designs supported at two points. The proof is in the appendix.

PROPOSITION 2.1. *The Bayesian D-optimal two-point design ξ_2^* with respect to the prior distribution π puts equal masses at the points*

$$(3) \quad t_1 = \frac{2k+1-\sqrt{2k+1}}{2E_{\pi_2}(\vartheta_2)}, \quad t_2 = \frac{2k+1+\sqrt{2k+1}}{2E_{\pi_2}(\vartheta_2)}$$

where $E_{\pi_2}[\cdot]$ denotes the expectation with respect to the marginal distribution π_2 of the prior π .

Setting $k = 0$ in (3) gives the best two-point designs obtained by Mukhopadhyay and Haines (1994) and Dette and Neugebauer (1994) for the simple exponential growth model. In the last-mentioned reference it is also proved that this design is Bayesian D-optimal for the exponential growth model if the gamma prior with density

$$(4) \quad \frac{\beta^\alpha \vartheta_2^{\alpha-1} \exp(-\beta \vartheta_2)}{\Gamma(\alpha)}, \quad \vartheta_2 > 0$$

is used for π_2 and the parameter α is sufficiently large, that is $\alpha \geq 3$. In the following we will demonstrate that similar results also hold for the more general model (1). To this end we assume that the parameter α in (4) satisfies $\alpha > 2k+1+\sqrt{2k+1}$ and assume in the proof of the following theorem without loss of generality for the scaling parameter $\beta = 1$. The general case can easily be obtained from this situation by multiplying the support points of the optimal designs for $\beta = 1$ with $\beta > 0$ and leaving the masses unchanged (note that $E_{\pi_2}(\vartheta_2) = \alpha/\beta$). By the equivalence theorem of

Whittle (1973) the design ξ_2^* in proposition 2.1 is optimal in the class of all designs if and only if

$$(5) \quad E_{\pi}[\text{tr}(I(\theta, \xi_2^*)^{-1} I(\theta, x))] = E_{\pi_2}[\text{tr}(I(\theta^*, \xi_2^*)^{-1} I(\theta^*, x))] \leq 2$$

for all $x \geq 0$ ($\theta^* = (1, \vartheta_2)'$), that is

$$(6) \quad F(x) = \frac{2x^{2k}}{(t_1 - t_2)^2} \left\{ \frac{(t_1 - x)^2}{t_2^{2k}(1 + 2(x - t_2))^\alpha} + \frac{(t_2 - x)^2}{t_1^{2k}(1 + 2(x - t_1))^\alpha} \right\} \leq 2$$

for all $x \geq 0$. Note that the integral in (5) exists because of the assumption $\alpha > 2k + 1 + \sqrt{2k+1}$. By the same reason we obtain $\lim_{x \rightarrow 0} F(x) = 0$ and obviously we have equality in (6) for $x = t_1$ and $x = t_2$. The following theorem shows that (5) (or equivalently (6)) is valid for all $x \geq 0$ if α is sufficiently large. A sketch of the proof can be found in the appendix.

THEOREM 2.2. Let (y_0, α_0) denote the unique solution of the system of equations

$$(7) \quad y(k+1 - \frac{\alpha}{2}) + \alpha(k+1 - \sqrt{2k+1}) - 2k^2 - k + k\sqrt{2k+1} + \frac{2\sqrt{2k+1}(\alpha-k)}{g(y, \alpha) + 1} = 0$$

$$\left(\frac{\alpha}{2} - (k+1)\right) - \frac{4(2k+1)(\alpha+1)(\alpha-k)}{(\alpha+y-2k-1)^2 - (2k+1)(g(y, \alpha) + 1)^2} = 0$$

where

$$g(y, \alpha) = \left[\frac{t_1}{t_2} \right]^{2k} \left[\frac{\alpha+y-2\alpha t_1}{\alpha+y-2\alpha t_2} \right]^{\alpha+1}$$

If $\alpha \geq \alpha_0$, then the design ξ_2^* defined in proposition 2.1 is Bayesian D-optimal with respect to the gamma prior (4) independently of β .

REMARK 2.3. Numerical calculations show that the condition $\alpha \geq \alpha_0$ is not necessary for the Bayesian D-optimality of the best two-point design with respect to the gamma prior (4). A simple necessary condition is that the second derivative of F at t_1 and t_2 is negative. Setting $F''(t_1) = 0$ gives the equation

$$(8) \quad -2k \log(\sqrt{2k+1} + 1) + (2k+1) \log(\sqrt{2k+1} - 1) + (\alpha+1) \log \alpha$$

$$- \alpha \log(\alpha - 2\sqrt{2k+1}) - \log\{\alpha(3\sqrt{2k+1} + 1) - (4k+2)(\sqrt{2k+1} - 1)\} = 0$$

and setting $F''(t_2) = 0$ yields

$$(9) \quad (2k+1) \log(\sqrt{2k+1} + 1) - 2k \log(\sqrt{2k+1} - 1) + (\alpha+1) \log \alpha \\ - \alpha \log(\alpha+2\sqrt{2k+1}) - \log\{\alpha(3\sqrt{2k+1} - 1) - (4k+2)(\sqrt{2k+1} + 1)\} = 0.$$

The α values satisfying these equations are not the same, for example for $k=2$ the equation (8) has a solution $\alpha = 12,4399$ while (9) has a solution $\alpha = 13,1167$. In fact the α -values satisfying (9) are larger than the corresponding values of (8) for all k , and thus provide an upper bound, α_1 , for the simple necessary condition, $\alpha > \alpha_1$. Some values for different parameters of α_0 and α_1 for various k are listed in table 2.1. The first row (α_0) shows the bounds for the sufficient condition while the second row (α_1) shows the corresponding values for the necessary condition (9). Finally, the third row gives the points α_2 for which the design with equal masses at the points given by (3) is Bayesian D-optimal with respect to the gamma prior (4) if and only if $\alpha \geq \alpha_2$. These α -values are obtained by solving the equations, $F(x) = 0$ and $F'(x) = 0$, simultaneously for α and x , and taking the solution for which x is not equal to t_1 and t_2 (cf. Mukhopadhyay & Haines (1994), example 4.1 with $b < 2/c$). Finally we remark that these bounds are valid for all parameters $\beta > 0$ of the prior distribution (4).

TABLE 2.1. Bounds for the parameter of the prior distribution (4). α_0 : sufficient condition, α_1 : necessary condition, α_2 : equivalent condition.

$k=$	0	1	2	3	4	5	...
α_0	2,5929	8,0233	13,2373	18,4397	23,6398	28,8395	...
α_1	2,5881	7,9810	13,1167	18,2274	23,3272	28,4207	...
α_2	2,5924	8,0179	13,2227	18,4089	23,5904	28,7702	...

Note that in the case $k = 0$ the necessary bound $\alpha_1 = 2.5581$ was also observed by Mukhopadhyay and Haines (1994) but the conjecture in the same reference that this bound is also sufficient for the existence of a Bayesian D-optimal design supported at two points is not correct. It is also remarkable that all three bounds $\alpha_0, \alpha_1, \alpha_2$ do not greatly differ. Therefore, for practical purposes, it seems to be adequate to verify only the necessary condition (9).

3. OPTIMAL DESIGNS WITH MORE THAN TWO SUPPORT POINTS

It is clear from the discussion of section 2 that for $\alpha < \alpha_2$ a two-point design cannot be Bayesian D-optimal with respect to the gamma prior (4). In this case the optimal designs have at least three support points and we have calculated the Bayesian D-optimal three-point designs using the numerical procedure "model 5.f" which is described in Sperlich (1993). The idea of this algorithm is to express the optimality criterion in terms of the coefficients appearing in the continued fraction expansion of the Stieltjes transform of a given design. The function Φ_π is now maximized in terms of these coefficients where the NAG routines D01FCF and E04KCF are used for the numerical integration and maximization (see Dette & Sperlich (1994) for more details). Some of the results for different parameters $k \geq 0$ are listed in table 3.1. The parameters α, β in the prior distribution (4) are chosen such that $E_{\pi_2}(\vartheta_2) = 1$ in all considered cases, that is $\alpha = \beta$.

The first row shows the support points of the Bayesian D-optimal three-point design while the second row contains the corresponding weights. Finally the efficiencies of the best two-point design given in theorem 2.2 with respect to the best three-point design can be found in the third row of the table. Here the efficiency is defined as the ratio of the optimality criterion $\Phi_\pi(\cdot)$ evaluated for the best two- and three-point design such that this ratio is in the interval $[0, 1]$. The Bayesian D-optimality of these designs within the class of all designs has been verified by inequality (5) of Whittle's equivalence theorem.

TABLE 3.1: Bayesian D-optimal three-point designs and the efficiency of the best two-point designs in the model (1) with respect to the best three-point design for various parameters k and different parameters $\alpha = \beta$ of the prior distribution (4). The designs marked with * are globally optimal.

$k = 0$	$\alpha = \beta = 1,5$	$\alpha = \beta = 2$	$\alpha = \beta = 2,5$
support	0,0000; 0,7656; 3,0320	0,0000; 0,8474; 2,2264	0,0000; 0,9574; 1,4842*
weights	0,4697; 0,3927; 0,1376	0,4856; 0,4073; 0,1071	0,4981; 0,4479; 0,0540
efficiency	0,9769	0,9952	0,9999

$k = 1$	$\alpha = \beta = 4$	$\alpha = \beta = 5$	$\alpha = \beta = 8$
support	0,5601; 1,7350; 3,7126	0,5833; 1,7980; 3,3426*	0,6337; 1,9091; 2,3705*
weights	0,4059; 0,3754; 0,2187	0,4330; 0,3426; 0,2244	0,4997; 0,0051; 0,4952
efficiency	0,9795	0,9910	1,0000

$k = 3$	$\alpha = \beta = 4$	$\alpha = \beta = 10$	$\alpha = \beta = 16$
support	1,7328; 3,6432; 7,7108	1,9827; 3,7703; 6,0288*	2,1312; 3,9554; 5,1471*
weights	0,3071; 0,4575; 0,2354	0,3917; 0,3509; 0,2574	0,4699; 0,1684; 0,3617
efficiency	0,5173	0,8884	0,9941

Some typical results are depicted in figure 1 for $k = 0$ and demonstrate that the best three-point designs are not globally optimal if $\alpha = \beta = 1,5$ and 2. It is remarkable that the answer of the question whether the three-point designs are Bayesian D-optimal in the class of all designs, depends again on the size of the parameters $\alpha = \beta$ in prior distribution (4). For $\alpha = \beta = 2,5$ the Bayesian D-optimal design is supported at three points and given in table 3.1 while this is not the case if $\alpha = \beta = 1,5$ and $\alpha = \beta = 2$. Here the Bayesian D-optimal design has at least four support points. It is also worthwhile to mention that this behaviour depends monotonically on the parameters $\alpha = \beta$. This means, if a Bayesian D-optimal design supported at at most three points exists for some parameter of the prior distribution (4), say $\alpha^* = \alpha = \beta$, then it also exists for all parameters $\alpha = \beta > \alpha^*$.

Bayesian D-optimal four-point designs in the case $k = 0$ are given in table 3.2 which also contains in its third row the efficiencies of the best two-point designs with respect to the four-point designs given in table 3.2. We observe only a minor improvement in the efficiency when one uses a four-point design instead of the three-point design.

TABLE 3.2: Bayesian D-optimal four-point designs in the model (1) for $k = 0$ and different parameters $\alpha = \beta$ of the prior distribution (4). The third row gives the efficiencies of the best two-point designs with respect to the four-point designs.

$k = 0$	$\alpha = \beta = 1,5$	$\alpha = \beta = 2$
support	0,0000; 0,6997; 1,9155; 7,2168	0,0000; 0,8456; 2,1878; 6,6996
weights	0,4679; 0,3313; 0,1711; 0,0297	0,4856; 0,4052; 0,1084; 0,0008
efficiency	0,9749	0,9951

All these observations correspond to the somewhat intuitive fact that the variance of the prior distribution (4) is a decreasing function of $\alpha = \beta$. Similar behaviour was also observed by Chaloner and Larntz (1989) in the case of logistic regression and discrete uniform prior distributions. Heuristically, a prior distribution with large variance requires more support points for the Bayesian D-optimal design than a distribution with smaller variance. If the variance of the prior distribution is 0, then the Bayesian optimality criterion reduces to the local D-optimality for the model (1). The local D-optimal design has equal weights at the two support points given by (3) for a one-point prior. This follows by the same arguments as in Mukhopadhyay and Haines (1994) for the case $k = 0$.

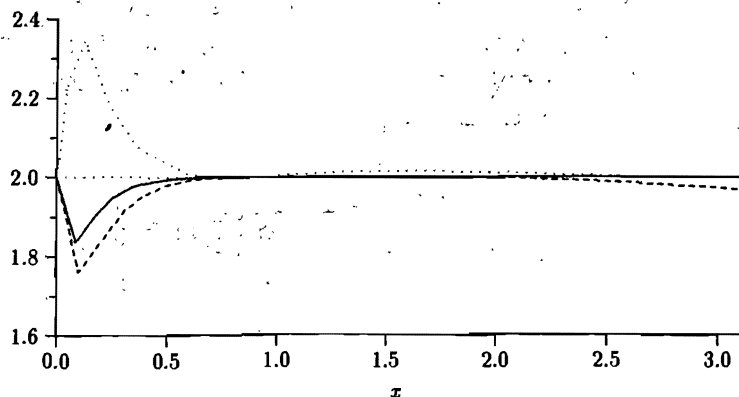


FIGURE 1: Condition (5) for the three-point designs in table 3.1 ($k = 0$). Solid line: $\alpha = \beta = 2.0$, dashed line: $\alpha = \beta = 2.5$, dotted line: $\alpha = \beta = 1.5$. For $\alpha = \beta = 2.5$ the best two-point design is globally optimal, in the two other cases the optimal design has at least four support points.

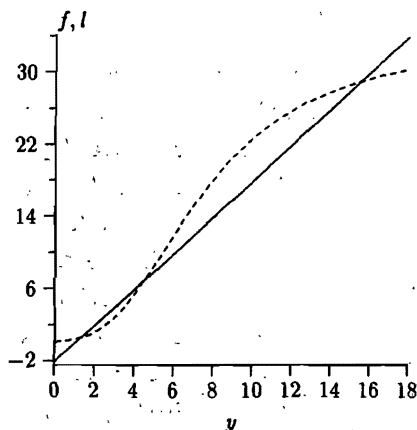


FIGURE 2: The functions $f(\cdot, \alpha)$ (dotted line) and $l(\cdot, \alpha)$ (solid line) for $k=2$, $\alpha=10$.

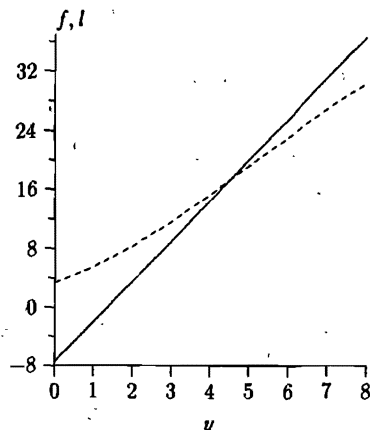


FIGURE 3: The functions $f(\cdot, \alpha)$ (dotted line) and $l(\cdot, \alpha)$ (solid line) for $k=2$, $\alpha=17$.

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4. APPENDIX: PROOF OF PROPOSITION 2.1 AND THEOREM 2.2

Proof of proposition 2.1. By standard arguments it follows that the best two-point design ξ_2^* must have equal masses at the two support points $t_1 < t_2$ (see e.g. Fedorov (1972), p. 85). Thus we obtain from (2) that

$$\Phi_{\pi}(\xi_2^*) = 2E_{\pi}[\log \vartheta_1] + E_{\pi_2}[-2\vartheta_2(t_1+t_2)] + 2k \log(t_1 t_2) + 2 \log(t_2 - t_1) - \log 4$$

which is maximized for t_1 and t_2 defined in (3). ■

Proof of theorem 2.2. The calculations used in the following proof are extremely tedious and for the sake of brevity we will only sketch the main lines of thought. Details can be found in Sperlich (1993). We assume $k \geq 0$ because the case $k = 0$ can be obtained by considering the limit $k \rightarrow 0$ from this case. An alternative proof for the case $k = 0$ can be found in Dette and Neugebauer (1994). These authors used Taylor expansions for the derivative of the function F in (6) in order to find the lower bound $\alpha^* = 3$, such that the two-point design of proposition 2.1 is globally Bayesian D-optimal whenever $\alpha \geq \alpha^*$. The present proof discusses the equivalence condition (5) more carefully and improves this bound to $\alpha_0 = 2,5929$ (see table 2.1). Let

$$f(y, \alpha) = \frac{2\sqrt{2k+1}(\alpha-k)}{g(y, \alpha) + 1}$$

and

$$l(y, \alpha) = y \left[\frac{\alpha}{2} - (k+1) \right] - \alpha \left[k+1 - \sqrt{2k+1} \right] + k \left[2k+1 - \sqrt{2k+1} \right].$$

From the form of the second derivative of f with respect to y it follows that there exists a number $y^* > 0$ such that for fixed α the function $f(\cdot, \alpha)$ is convex on the interval $[0, y^*)$ and concave on the interval $[y^*, \infty)$. By a consideration of the monotonicity properties of the functions $f(y, \cdot)$ and $l(y, \cdot)$ for fixed y it can now be shown that there exists a number α_0 such that the following holds true.

- (1) If $\alpha < \alpha_0$, then the functions $f(\cdot, \alpha)$ and $l(\cdot, \alpha)$ have exactly three points of intersection (see figure 2).
- (2) If $\alpha > \alpha_0$, then the functions $f(\cdot, \alpha)$ and $l(\cdot, \alpha)$ have exactly one point of intersection (see figure 3).

- (3) If $\alpha = \alpha_0$, then the functions $f(\cdot, \alpha)$ and $l(\cdot, \alpha)$ have exactly two points of intersection (see figure 4). At one of these points $l(\cdot, \alpha_0)$ is the tangent of $f(\cdot, \alpha_0)$.

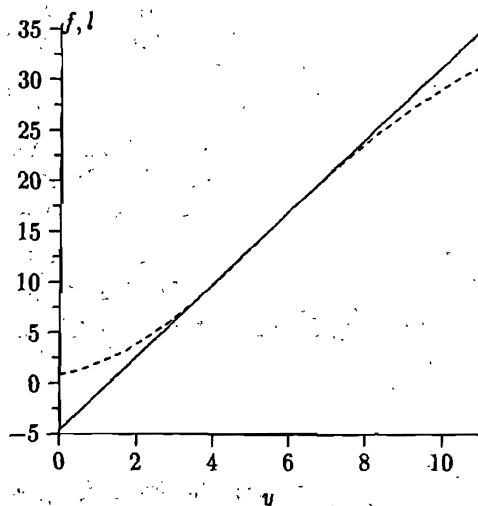


FIGURE 4 The function $h(\cdot, \alpha)$ (solid line) and $f(\cdot, \alpha)$ (dotted line) for $k=2$, $\alpha = 13.2373 = \alpha_0$

It is easy to see that the two equations in (7) are equivalent to the system

$$l(y, \alpha) = f(y, \alpha), \quad \frac{\alpha}{2} - (k+1) = \frac{\partial}{\partial y} l(y, \alpha) = \frac{\partial}{\partial y} f(y, \alpha)$$

and by the preceding discussion there exists exactly one solution of (7), namely the pair (y_0, α_0) such that the line $l(\cdot, \alpha_0)$ is the tangent to $f(\cdot, \alpha_0)$ at the point y_0 . In order to show the inequality (6), which implies the assertion of the theorem, we now calculate from (6)

$$F'(x) = \frac{4x^{2k-1}(t_2-x)(t_1-x)}{\alpha(t_1-t_2)^2} \left[\frac{2\sqrt{2k+1}(\alpha-k) - l(2\alpha x, \alpha)}{t_1^{2k}(1+2(x-t_1))^{\alpha+1}} - \frac{l(2\alpha x, \alpha)}{t_2^{2k}(1+2(x-t_2))^{\alpha+1}} \right]$$

which can be rewritten as

$$(10) \quad F'(x) = \frac{4x^{2k-1}(t_2-x)(t_1-x)}{\alpha(t_1-t_2)^2 t_1^{2k}(1+2(x-t_1))^{\alpha+1}} [g(2\alpha x, \alpha) + 1] [f(2\alpha x, \alpha) - l(2\alpha x, \alpha)].$$

Obviously we have $F'(t_1) = F'(t_2) = 0$ and $F(t_1) = F(t_2) = 2$. In the case (2) above, $\alpha > \alpha_0$, it follows that the function $h(x) = f(2\alpha x, \alpha) - l(2\alpha x, \alpha)$ has exactly one zero $t^* \in (t_1, t_2)$ and from $h(0) > 0$, $\lim_{y \rightarrow \infty} h(y) = -\infty$, that F has a local minimum at $t^* \in (t_1, t_2)$. Because of $\lim_{x \rightarrow \infty} F(x) = 0$ and the fact that F' has no further zeroes the inequality (6) holds for all $x \geq 0$. In case (3), $\alpha = \alpha_0$, the function h has at most two zeros. A point $t_1^* \in (t_1, t_2)$, where the sign of h is changing, is always a zero but there could also exist another zero of h , say $t_2^* \geq t_2$, where the sign of h is not changing. Consequently F is decreasing for all $x \geq t_2$ and (6) follows by the same reasoning as in the case $\alpha > \alpha_0$. ■

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